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[23ESME12]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

ELEMENTS OF MECHANICAL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Missing data may be suitably assumed.
ii. Answer all questions.

		PART - A	**05 X 02=10**		
1.	**a**	Why lubrication is necessary in the bearings?	L2	CO3	**02**
	b.	Provide the GD & T symbols for the following **i.** Flatness **ii.** Cylindricity	L2	CO4	**02**
	c.	How gears are classified?	L2	CO2	**02**
	d.	List the sources of energy.	L2	CO1	**02**
	e.	How knurling operation is performed on a lathe machine?	L2	CO5	**02**
		PART - B	**06 X 05=30**		
2.	**a.**	Explain horizontal axis wind turbine.	L2	CO1	**05**
		OR			
	b.	What is priming? Why it is necessary in pumps?	L2	CO1	**05**
3.	**a.**	State the difference between tempered and laminated glasses	L2	CO2	**05**
		OR			
	b.	Write a short note on magnesium and titanium alloys?	L2	CO2	**05**
4.	**a.**	Explain clearance fits and interference fits with suitable examples.	L2	CO4	**05**
		OR			
	b.	What do you understand by the process of Caulking and Fullering	L2	CO4	**05**
5.	**a.**	List the conditions for using helical compression spring.	L2	CO3	**05**
		OR			
	b.	Sketch a typical milling machine and label its parts.	L2	CO5	**05**
6.	**a.**	Differentiate between rolling contact bearing and sliding contact bearings.	L2	CO3	**05**
		OR			
	b.	Explain the working of solar power plant	L2	CO1	**05**
7.	**a.**	Discuss the mechanical properties of the materials in details	L2	CO3	**05**
		OR			
	b.	Explain different operations performed on milling machine with neat sketches.	L2	CO5	**05**
		PART - C	**06X10=60**		
8.	**a.**	With a neat sketch, explain hydroelectric power plant.	L2	CO1	**10**
		OR			

	b.	Explain the working principle and operation of a centrifugal pump.	L2	CO1	**10**
9.	**a.**	Explain the properties of steels and alloys of aluminum.	L2	CO2	**10**
		OR			
	b.	What do you mean by tests for aircraft grade glass? Explain any two testing methods.	L2	CO2	**10**
10.	**a.**	Explain compression springs and torsion springs with neat sketches.	L2	CO3	**10**
		OR			
	b.	Explain cylindrical roller bearing, tapered roller bearing with neat sketch.	L2	CO3	**10**
11.	**a.**	Discuss the strength analysis of welded joints and riveted joint.	L2	CO4	**10**
		OR			
	b.	Define the process of welding. Explain its types	L2	CO4	**10**
12.	**a.**	A Pair of gear has been designed with velocity ratio of 3.20. The pinion has 20 teeth and circular pitch of 78.54mm. **i.** Determine number of teeth on gear **ii.** Calculate module of gear **iii.** Center Distance	L2	CO3	**10**
		OR			
	b.	What operation are carried on lathe machine and explain in brief with neat sketch?	L2	CO5	**10**
13.	**a.**	Explain the parts of the lathe with a neat labeled diagram.	L2	CO5	**10**
		OR			
	b.	Explain the construction of shaping machine and operation carried on it.	L2	CO5	**10**